

Wireless Network Traffic Forecasting Competition—Data Description

1. Dataset Overview

This competition provides **uplink** network traffic time series at **two spatial resolutions**:

- **Cell level:** 30 anonymized wireless network cells;
- **Grid level:** 273 anonymized spatial grids subordinate to those 30 cells.

Traffic is provided as the **sum** within each 5-minute window, with a sampling interval of **5 minutes**. Both levels share the same regular time axis: the data continuously cover **7 days + 5 minutes**, and each series contains **2,017** time points.

A full day contains:

$$24 \times 60 \div 5 = 288$$

time points. Therefore, the observation length for each series is:

$$288 \times 7 + 1 = 2,017$$

The time range is from 2024-11-30 23:50:00 to 2024-12-07 23:50:00, inclusive. The two endpoints are exactly 7 days apart; including both endpoints gives 2,017 time points per series, corresponding to a total coverage of 7 days + 5 minutes.

Cell-level and grid-level tables are temporally aligned. Most cells have 10 subordinate grids; three cells have only 1 subordinate grid in the released data ($27 \text{ cells} \times 10 + 3 \text{ cells} \times 1 = 273$ grids). Participants may use hierarchical structure, cross-series correlation, and timestamp-derived features for forecasting. Cell-level traffic and the sum of a cell's subordinate grids are **not** required to be identical; they should be treated as two related but separately measured datasets.

2. Basic Dataset Information

Attribute	Description
Number of wireless network cells	30
Number of subordinate grids	273
Time points per series	2,017
Dataset duration	7 days + 5 minutes
Temporal resolution	5 minutes
Samples per hour	12
Samples per day	288
Cell-level observations	$30 \times 2,017 = 60,510$
Grid-level observations	$273 \times 2,017 = 550,641$
Missing timestamps	0
Start time	2024-11-30 23:50:00 (Saturday)
End time	2024-12-07 23:50:00 (Saturday)
Time zone	Based on the recorded collection time; no separate time zone is specified
Cell matrix shape	(T=2017, N=30), non-negative integers
Grid matrix shape	(T=2017, N=273), non-negative integers

Each record represents the sum of uplink traffic for the corresponding cell or grid within a **5-minute aggregation window**.

The **timestamp** indicates the **start time of the aggregation window**. A timestamp t corresponds to the interval: $[t, t + 5 \text{ min})$

For example, 2024-11-30 23:50:00 corresponds to $[2024-11-30 \text{ 23:50:00}, 2024-11-30 \text{ 23:55:00})$.

The timestamps follow the actual collection calendar, covering Saturday, November 30, 2024 at 23:50 through Saturday, December 7, 2024 at 23:50. Participants may use timestamp-derived features such as day of week and weekend indicators. Because each day of the week appears only once, weekday effects should not be overfit.

3. Network Traffic Definition

The target variable represents the **summed uplink traffic count** for the corresponding wireless network cell or spatial grid within a specified 5-minute window.

Attribute	Description
Metric name	Uplink Traffic
Aggregation window	5 minutes
Metric meaning	Total uplink traffic for the cell/grid within the window
Aggregation method	Sum of uplink traffic within each 5-minute window
Data unit	Anonymized raw count unit
Traffic direction	Uplink traffic
Numeric transformation	No normalization, standardization, or logarithmic transformation
Cell value type	Non-negative integer; minimum 2,088 and maximum 28,952,521
Grid value type	Non-negative integer; minimum 0 and maximum 27,701,840

3.1 Cell-Level Distribution

Observed distribution across all 60,510 cell-level records:

Statistic	Value
Minimum	2,088
1% quantile	52,883
25% quantile	556,311
Median	1,115,156
Mean	1,355,193
75% quantile	1,825,769
99% quantile	5,088,095
Maximum	28,952,521
Proportion with traffic $\geq 3 \times 10^5$	85.35%
Proportion with traffic $\geq 10^6$	55.11%

Mean traffic levels vary across cells: the median of the per-cell means is approximately 1.25×10^6 , with the lowest at about 6.39×10^5 and the highest at about 2.35×10^6 .

3.2 Grid-Level Distribution

Observed distribution across all 550,641 grid-level records (after missing-value imputation described in Section 8):

Statistic	Value
Minimum	0
1% quantile	0
25% quantile	1,446
Median	10,686
Mean	530,611
75% quantile	45,189
99% quantile	9,247,470
Maximum	27,701,840
Proportion with traffic = 0	7.98%
Proportion with traffic $\geq 10^4$	51.14%
Proportion with traffic $\geq 10^5$	17.16%

Grid means are highly heterogeneous: the median of the per-grid means is approximately 2.50×10^4 , with the lowest at about 7.20×10^2 and the highest at about 9.19×10^6 .

4. Data Entities

4.1 Cells

The cell-level dataset contains **30 wireless network cells**, corresponding one-to-one with columns `Cell_001`–`Cell_030`.

cell_id	Column index	Meaning
Cell_001	0	First anonymized wireless network cell
Cell_002	1	Second anonymized wireless network cell
...	...	...
Cell_030	29	30th anonymized wireless network cell

4.2 Grids

The grid-level dataset contains **273 spatial grids** subordinate to the 30 cells. Each grid is assigned a unique anonymized identifier `Grid_001`–`Grid_273`. Membership is provided in `grid_cell_membership.csv`:

Field	Type	Description
grid_id	string	Anonymized grid identifier
cell_id	string	Parent anonymized cell identifier
grid_index_within_cell	integer	Local grid index within the parent cell

Item	Value
Cells with 10 grids	27
Cells with 1 grid	3
Total grids	273

To protect information about the actual communications network, the true cell/grid names, physical locations, base-station identifiers, and any other information that could identify real network locations have been **anonymized and are not disclosed to participants**.

A given `cell_id` or `grid_id` always refers to the same entity, so historical observations from the same series may be used for modeling. Parent–child membership between cells and grids is stable over time.

5. Data Fields

Both levels use a **wide-table format**: each row is a timestamp and each series is represented by one column.

5.1 Cell-Level Fields

Field	Type	Description
timestamp	datetime	Start time of the current 5-minute aggregation window, in YYYY-MM-DD HH:MM:SS format; sorted in ascending time order
Cell_001 ... Cell_030	integer	Summed uplink traffic count for the corresponding cell within the time window

Example:

```
timestamp,Cell_001,Cell_002,...,Cell_030
2024-11-30 23:50:00,1859060,1265545,...,<value>
2024-11-30 23:55:00,1947749,3346731,...,<value>
2024-12-01 00:00:00,2435073,3116848,...,<value>
```

5.2 Grid-Level Fields

Field	Type	Description
timestamp	datetime	Same definition as the cell-level table
Grid_001 ... Grid_273	integer	Summed uplink traffic count for the corresponding grid within the time window

Example:

```
timestamp,Grid_001,Grid_002,...,Grid_273
2024-11-30 23:50:00,<value>,<value>,...,<value>
2024-11-30 23:55:00,<value>,<value>,...,<value>
2024-12-01 00:00:00,<value>,<value>,...,<value>
```

Reading a column from top to bottom gives the complete traffic time series for that cell or grid. Reading a row from left to right gives the cross-sectional traffic values of all series at the same timestamp.

6. Data Files

The following files are provided to participants (see the `competition_data/public/` package).

6.1 Cell-level files

cell_train.csv

Item	Value
Time range	2024-11-30 23:50:00 – 2024-12-06 23:45:00
Rows (time points)	1,728 (exactly 6 days)
Number of cell columns	30
Traffic observations	51,840

cell_test.csv The test file provides the timestamps to be predicted and the same cell columns as the training set. The cell columns are empty and contain **no** ground-truth traffic values.

Item	Value
Time range	2024-12-06 23:50:00 – 2024-12-07 23:50:00
Rows (time points)	289 (1 day + 5 minutes)
Number of cell columns	30
Entries to predict	8,670

cell_sample_submission.csv The sample submission has the same format as the scoring file: the first column is `timestamp`, followed by predicted values for `Cell_001–Cell_030`.

```
timestamp,Cell_001,Cell_002,...,Cell_030
2024-12-06 23:50:00,<prediction>,<prediction>,...,<prediction>
2024-12-06 23:55:00,<prediction>,<prediction>,...,<prediction>
...
```

6.2 Grid-level files

grid_train.csv

Item	Value
Time range	2024-11-30 23:50:00 – 2024-12-06 23:45:00
Rows (time points)	1,728 (exactly 6 days)
Number of grid columns	273
Traffic observations	471,744

`grid_test.csv`

Item	Value
Time range	2024-12-06 23:50:00 – 2024-12-07 23:50:00
Rows (time points)	289 (1 day + 5 minutes)
Number of grid columns	273
Entries to predict	78,897

`grid_sample_submission.csv` Same format as `grid_test.csv`, with numeric predictions for Grid_001–Grid_273.

6.3 Membership file

`grid_cell_membership.csv` maps each `grid_id` to its parent `cell_id`. This file contains no traffic values and may be used freely for hierarchical modeling.

Values in the sample submission files are generated by a **seasonal naive baseline** (traffic from the same clock time one day earlier in the training segment) and are provided only to demonstrate the required format; they are not reference answers.

Submission checklist for each level:

- Include every timestamp in the corresponding test file and all series columns, with no missing or additional rows or columns.
- Prefer keeping the same row order and column order as the test file.
- Each series column must contain numeric values; predictions may be floating-point numbers.

7. Training/Test Split

The dataset is split **chronologically**; no random sampling is performed across series or time points. The same split is used for both cell-level and grid-level tracks.

Training Time Range

2024-11-30 23:50:00 -- 2024-12-06 23:45:00
(windows 0 through 1727, 1,728 points / 6 days in total)

Test Time Range

2024-12-06 23:50:00 -- 2024-12-07 23:50:00
(windows 1728 through 2016, 289 points / 24 hours 5 minutes in total)

Models may use only information that has already been provided before each test time. Test-segment labels are unavailable before evaluation. This is a **multi-series, multi-step direct forecasting** task at each spatial level: forecast the next 289 consecutive 5-minute points for all 30 cells and/or all 273 grids.

Permitted information includes:

- Historical traffic from the same cell or grid;
- Hierarchical cell–grid membership;
- Timestamp-derived features such as temporal cycles and intraday position;
- Correlations across different cells and/or grids;
- Other features constructed solely from the publicly provided training data.

The ground-truth traffic values in the test segment and any non-public external operational data are not permitted.

8. Data Completeness and Anomalies

8.1 Missing Values

Cell level. All 30 cells have records at all 2,017 time points. There are **no** NaN values, infinite values, or missing timestamps.

Grid level. The raw grid extracts contained 29,229 missing entries (5.31% of $273 \times 2,017$). In the released competition tables these missing entries have been imputed as **0**. After imputation there are no NaN values left. The time axis remains a strictly regular 5-minute grid.

8.2 Zero Values

Cell level. There are **no** observations with traffic = 0. All cell observations satisfy traffic $\geq 2,088$.

Grid level. About 7.98% of released grid observations are exactly 0 (including imputed missing entries and genuine zeros in the source extracts). Zeros should be handled carefully under percentage-error metrics.

8.3 Outliers and Level Shifts

Both levels contain short-term peaks (cell maximum approximately 2.90×10^7 ; grid maximum approximately 2.77×10^7). Such bursts may reflect genuine changes in network activity and should not automatically be treated as data-entry errors.

A small number of cell series exhibit day-to-day level shifts (the maximum day-to-day change in the log daily mean can exceed 1.0). These series are retained. Participants may choose whether to apply anomaly detection, segmentation, or robust processing.

8.4 Intraday Cycle

According to the collection clock, average cell-level traffic across the full dataset is lowest at approximately 05:00 and highest at approximately 22:00. Grid-level averages show a similar trough near 05:00 and peak near 22:00. The strength and phase of the cycle are not identical across series.

9. Data Anonymization and Privacy Protection

The dataset has been anonymized:

- It contains no user identity information;
- True geographic locations, base-station names, and other information that could identify network infrastructure are not provided;
- `cell_id` and `grid_id` are used only to distinguish entities and carry no geographic or business meaning;
- Timestamps retain the actual collection calendar, but locations are not provided, so specific network infrastructure cannot be identified from the timestamps;
- Only the hierarchical membership between anonymized cells and grids is retained.

Traffic values retain the raw counts from the source data (after grid missing-value imputation) and have **not** undergone any additional normalization or scaling. The values cannot be directly converted into real physical traffic units.

10. Dataset Characteristics

This dataset is a **high-frequency, multi-resolution, multi-series wireless network traffic time-series dataset**.

10.1 High Temporal Resolution

The temporal resolution is 5 minutes, giving 288 points per series per day and capturing fine-grained changes in network activity over time.

10.2 Dual Spatial Resolution

Traffic is released at both cell and subordinate-grid resolutions with a shared time axis, enabling hierarchical, multi-scale, and transfer-style modeling.

10.3 Multiple Time Series

The dataset covers 30 cells and 273 grids simultaneously. Traffic scale, periodicity, and volatility vary substantially across series.

10.4 Intraday Periodicity

Traffic follows the clock, with troughs in the early morning and peaks in the evening. The primary period is 288 time points (1 day).

10.5 Cross-Series Heterogeneity

Mean traffic levels can differ by orders of magnitude across grids, so a single set of hyperparameters may not be optimal for every series.

10.6 Potential Cross-Series Correlation

Different cells and grids may change synchronously during the same time periods and may therefore be modeled jointly, including through parent-child cell-grid structure.

10.7 Short History

Each series has only about 7 days of history. Long-period effects such as weekly patterns cannot be estimated reliably from the data, so models should rely primarily on intraday periodicity, recent traffic levels, hierarchy, and cross-series information.

11. Submission Requirements

Participants may submit predictions for the cell-level track, the grid-level track, or both. Each submitted file must satisfy:

1. **Cell track:** first column `timestamp`, followed by `Cell_001` through `Cell_030`; column names and order must match `cell_test.csv`.
2. **Grid track:** first column `timestamp`, followed by `Grid_001` through `Grid_273`; column names and order must match `grid_test.csv`.
3. The set of rows must correspond one-to-one with the `timestamp` values in the corresponding test file, with no missing or additional rows.
4. A prediction must be provided for every timestamp in every series column.
5. Prediction values must be numeric.

Non-negative predictions are recommended. If any non-numeric or missing entry is found during evaluation, the submission will be considered invalid.

12. Evaluation Metrics

This competition uses **two spatial tracks**. Within each track, **5 official metrics** are ranked independently and each awards one winner. Participants need to submit only one prediction file per track; the platform computes all metrics separately on the same set of predictions.

12.1 Cell-Level Track

Let all cell test-set entries be $\{(y_i, \hat{y}_i)\}_{i=1}^n$, where $n = 8,670$. Masked MAPE is calculated only on entries whose ground-truth values meet the specified threshold:

Track	Metric	Definition	Evaluation Scope
1	MAE	$\frac{1}{n} \sum_i \hat{y}_i - y_i $	All cell test-set entries
2	RMSE	$\sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2}$	All cell test-set entries
3	MAPE	$\frac{100}{n} \sum_i \hat{y}_i - y_i / y_i$	All cell test-set entries
4	Masked MAPE @ 3e5	$\frac{100}{n'} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 3 \times 10^5$ (approximately 7,268 entries, 83.8%)
5	Masked MAPE @ 1e6	$\frac{100}{n'} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 10^6$ (approximately 4,742 entries, 54.7%)

12.2 Grid-Level Track

Let all grid test-set entries be $\{(y_i, \hat{y}_i)\}_{i=1}^n$, where $n = 78,897$. Because grid traffic is more sparse and can be zero, masked MAPE uses lower thresholds:

Track	Metric	Definition	Evaluation Scope
1	MAE	$\frac{1}{n} \sum_i \hat{y}_i - y_i $	All grid test-set entries
2	RMSE	$\sqrt{\frac{1}{n} \sum_i (\hat{y}_i - y_i)^2}$	All grid test-set entries
3	MAPE	$\frac{100}{n} \sum_i \hat{y}_i - y_i / y_i$	Entries with $y_i > 0$ only
4	Masked MAPE @ 1e4	$\frac{100}{n'} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 10^4$ (approximately 40,231 entries, 51.0%)
5	Masked MAPE @ 1e5	$\frac{100}{n'} \sum_i \hat{y}_i - y_i / y_i$	$y_i \geq 10^5$ (approximately 13,656 entries, 17.3%)

For all metrics, **lower is better**. Each metric within each spatial track has an independent winner. If two valid submissions have exactly the same value on the same metric, the earlier valid submission wins the tie. The winners of different metrics or tracks may be the same team or different teams.

The mask for Masked MAPE is determined solely by the **ground-truth** value y_i , independently of the prediction. Entries outside the threshold are excluded from the average for that metric.

13. Dataset Size Summary

Metric	Value
Number of wireless network cells	30
Number of subordinate grids	273
Temporal resolution	5 minutes
Samples per hour	12
Samples per day	288
Dataset duration	7 days + 5 minutes
Observations per series	2,017
Cell-level total observations	60,510
Grid-level total observations	550,641
Training rows	1,728
Test rows	289
Cell training observations	51,840
Cell test entries to predict	8,670
Grid training observations	471,744
Grid test entries to predict	78,897
Forecast steps	289